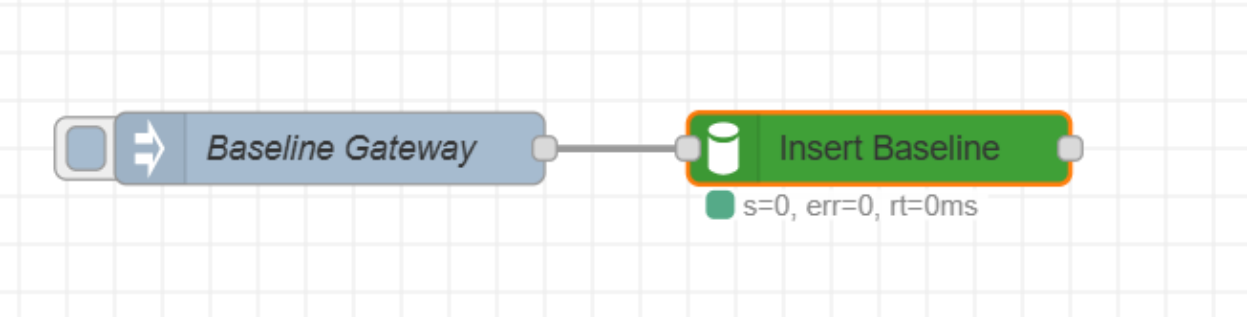
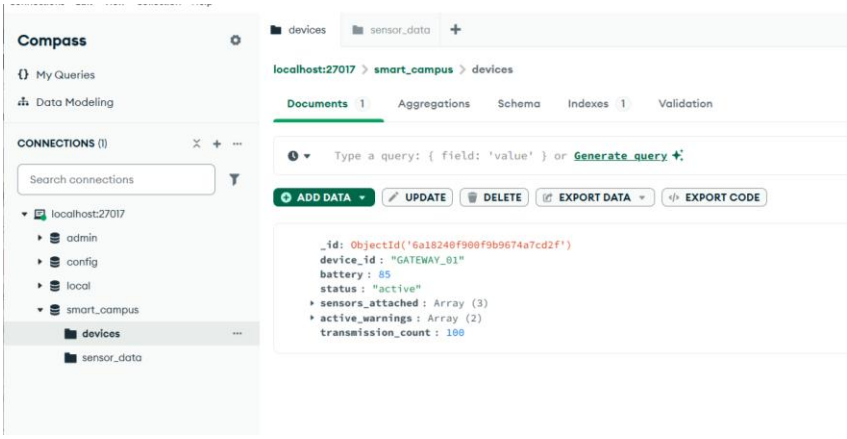
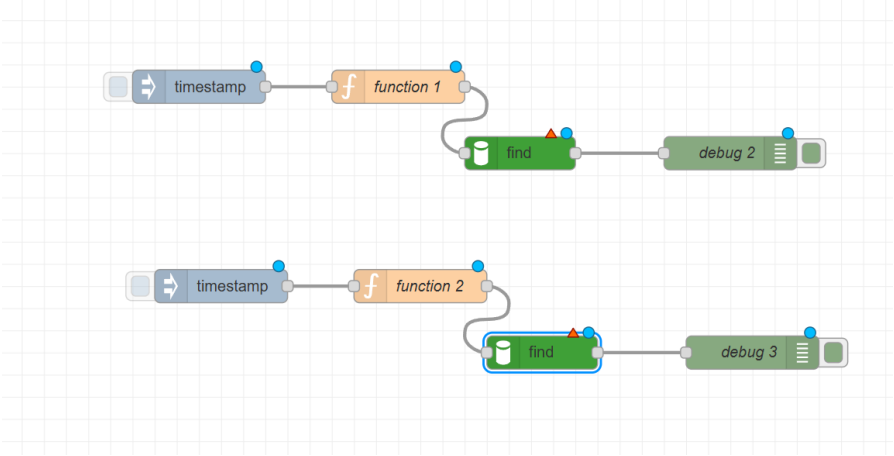
Exercise 7: Advanced NoSQL Query Operators

Step 1: Baseline Data Initialization

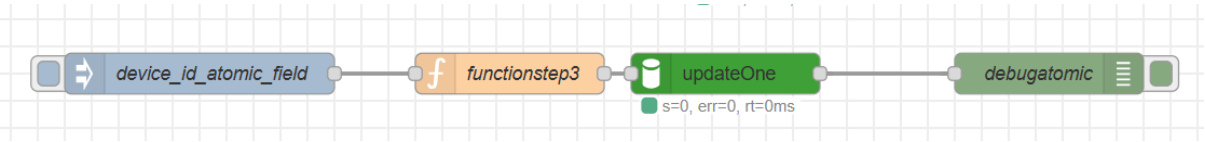


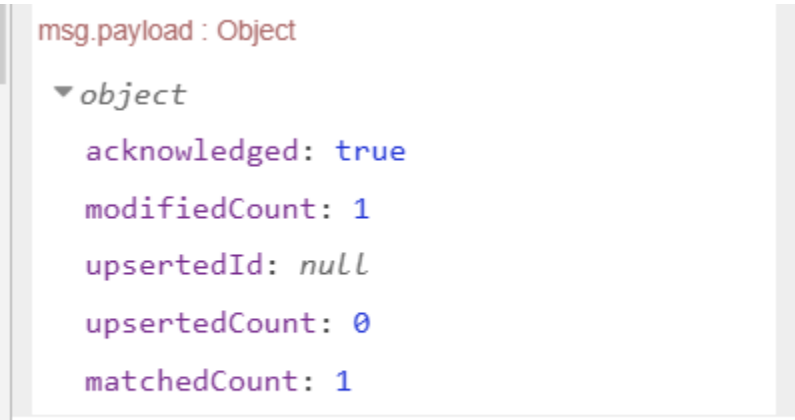


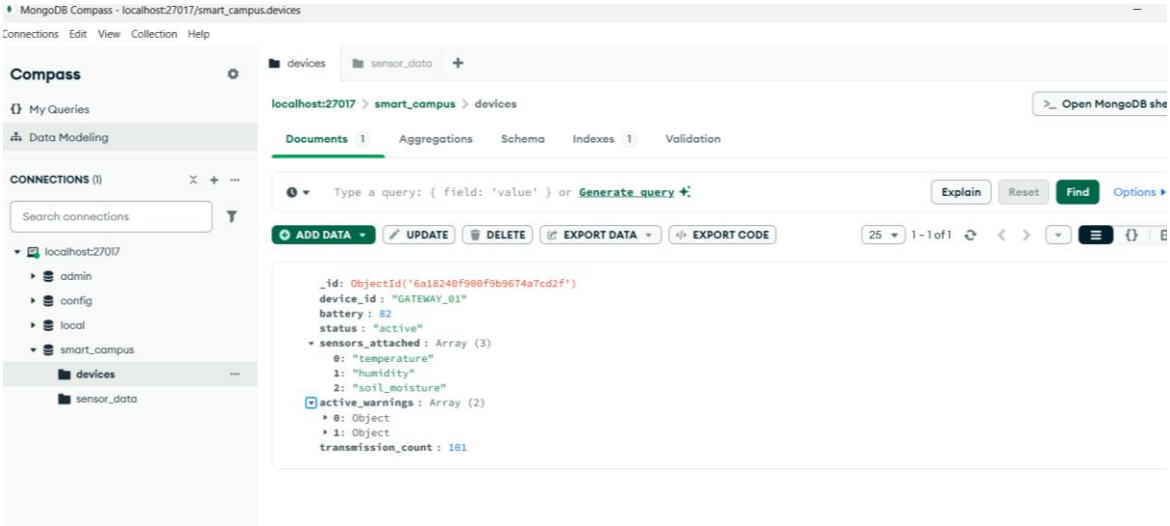
Step 2: Comparison and Logical Operators



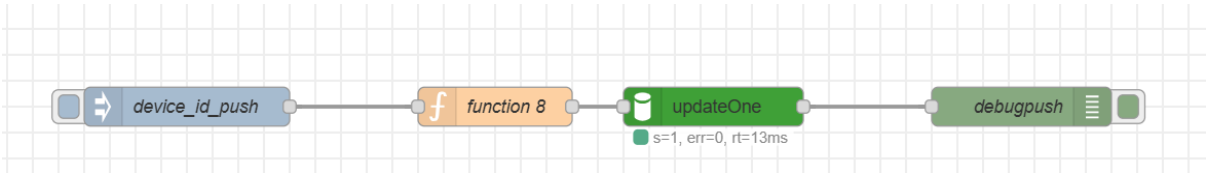
Step 3: Atomic Field Modifiers

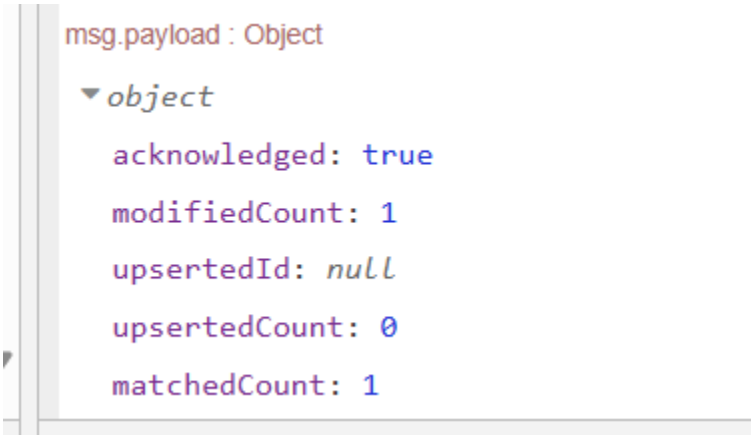


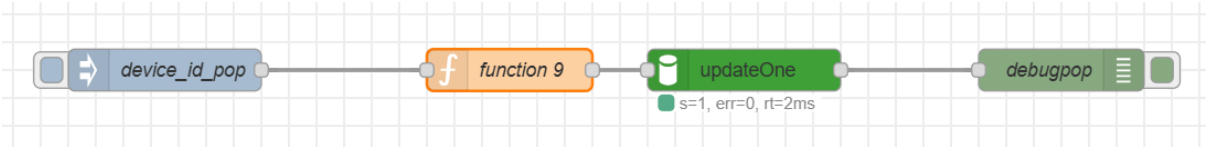


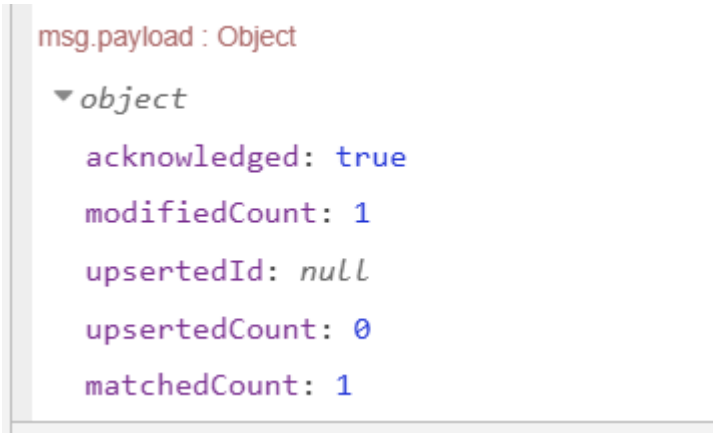


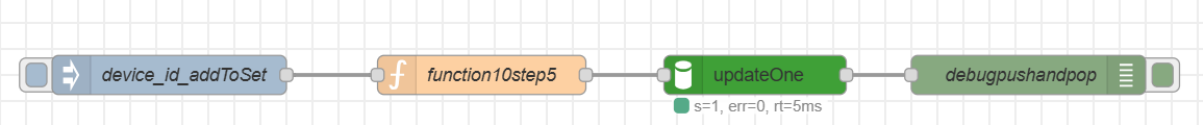
Step 4: Complex Array Manipulation

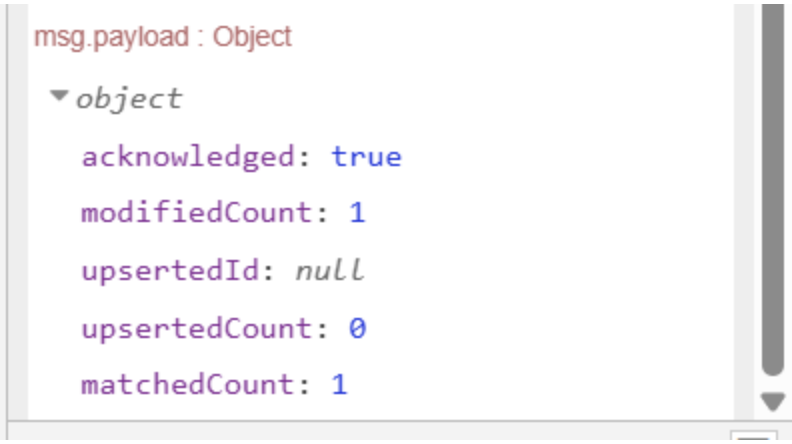












Part 5: Discussion & Architectural Analysis

Question 1: The $inc vs. $set Paradigm

The $inc vs. $set Paradigm: Using native atomic operations avoids **lost update anomalies** (race conditions) caused by the latency between reading data into Node-RED and writing it back. Native modifiers handle calculations inside a single transaction pass at the storage level.

Question 2: Efficient Payload Management

Efficient Payload Management: Utilizing native array modifiers (like $push or $addToSet) significantly optimizes network performance. Instead of downloading an entire array over the network, mutating it in Node-RED's memory, and re-uploading it, Node-RED only sends a tiny instructional delta string. The database engine does the heavy lifting locally.